Design and Evaluation of a Reproducible Machine Learning Workflow for Multiclass Yeast Protein Localization Prediction

Mustafa Barkin Kemec

Abstract

Predicting the cellular localization of proteins is a significant problem in bioinformatics, as protein function is closely related to its localization within the cell. The project involved exploratory data analysis, feature selection, model evaluation, algorithm comparison, and hyperparameter optimization. Multiple machine learning models were evaluated, including Logistic Regression, k-Nearest Neighbors, Decision Trees, Naive Bayes, Support Vector Machines, Random Forest, and ExtraTrees. Due to the highly imbalanced and nonlinear nature of the dataset, ensemble tree-based methods achieved the best overall performance. Random Forest was selected as the final model and further optimized using GridSearchCV. The optimized model achieved a reference accuracy rate of 62% with a Matthews Correlation Coefficient (MCC) of 0.51. The results showed significant improvements compared to the basic Logistic Regression model, especially for minority localization classes. In conclusion, this demonstrates that ensemble-based machine learning methods are better suited for modeling complex protein localization patterns compared to simpler linear approaches.

All analysis and documentation are available on GitHub.

Introduction

Protein localization is the mechanism by which cells transport and maintain proteins at specific locations to ensure proper cellular function (Hung et al, 2011). It is a fundamental aspect of cellular organization and biological function. Localization errors can disrupt essential cellular processes such as metabolism, signal transduction, transport, and gene regulation. Therefore, predicting the subcellular localization of proteins represents a significant challenge in molecular biology and bioinformatics.

Experimental techniques used to determine protein localization, including fluorescence microscopy and biochemical fractionation, are often both time-consuming and expensive when applied on a large scale. As biological databases continue to grow rapidly, computational approaches to support large-scale protein annotations have become increasingly important. In particular, machine learning methods provide effective tools for identifying complex patterns within biological data and predicting localization regions based on sequence-derived features.

The yeast, *Saccharomyces cerevisiae*, is widely used as a model organism in molecular and cellular biology due to its relatively simple eukaryotic structure and well-characterized genome (Botstein et al., 2011). The yeast dataset from the UCI Machine Learning Repository (Nakai, K., 1991) contains numerical features that define proteins and known localization regions, and providing a suitable benchmark for multi-class classification tasks in bioinformatics.

In this study, a machine learning pipeline was developed to predict protein localization regions using the UCI yeast dataset. The workflow included data analysis, feature evaluation, class-sensitive data splitting, model training, and performance evaluation using multiple evaluation metrics. Special attention was paid to reproducibility and leakage-sensitive evaluation procedures during feature selection and model validation. The aim of this study was not only to compare prediction performance among machine learning models but also to investigate the biological significance of selected features and the challenges associated with multi-class and imbalanced biological datasets.

2. Materials and Methods

2.1 Dataset

The dataset used in this study is the Yeast Protein Localization dataset obtained from the UCI Machine Learning Repository. It contains a total of 1,484 protein samples.

Each sample is represented by eight numerical features that define different sequence-derived and physicochemical properties of proteins: mcg, gvh, alm, mit, erl, pox, vac, and nuc. The target variable corresponds to the protein localization region, creating a multi-class classification problem involving ten different cellular compartments.

The dataset presents several features commonly encountered in biological machine learning tasks, including class imbalance and overlapping feature distributions between localization classes. Some localization regions contain significantly fewer samples than others, increasing the difficulty of model generalization and evaluation. Furthermore, no data completion operations were required as there were no missing values in the dataset.

2.2 Feature Definition

The Yeast dataset contains eight numerical features. These features are derived from sequence analysis.

The features used in this study are summarized below (Nakai, K., 1991):

mcg: McGeoch method for signal sequence recognition.

gvh: Score derived from the von Heijne method for signal sequence analysis.

alm: Score of amino acid composition in the N-terminal region.

mit: Prediction score related to mitochondrial localization.

erl: Presence of endoplasmic reticulum retention motifs.

pox: Score for peroxisomal targeting.

vac: Score for vacuolar localization signals.

nuc: Score for nuclear localization signals.

All feature values are numerical and have already been normalized within comparable ranges (between 0-1) in the original dataset.

2.3 Data Preparation and Fold Construction

Before model development, the data was divided into folds to ensure reproducibility and prevent data leakage during evaluation.

Instead of a simple random training-test split, a special fold-based split was performed. Because class imbalance exists, the data was divided into a total of 4 folds, paying attention to class distribution. One-fold was set aside as an independent benchmark set and labeled with the '-1' fold value, while the remaining three folds (0, 1, and 2) were used for cross-validation and training during model development and feature selection.

This strategy was adopted to separate the final benchmark evaluation from the model optimization process. Specifically, to prevent data leakage and overly optimistic performance predictions, the benchmark fold was completely excluded from the feature selection and model tuning procedures.

The resulting workflow provided a suitable, reproducible, and leakage-sensitive evaluation framework for multi-class biological classification problems.

2.4 Exploratory Data Analysis

Exploratory data analysis was conducted prior to model development to examine class distribution, feature distributions, feature correlations, and potential discrimination patterns between localization classes, and to facilitate biological interpretation.

Various graphs were generated, including class distribution plots, histograms, density plots, box plots, correlation heat maps, and Principal Component Analysis (PCA). These analyses were used to identify potential problems such as class imbalance, skewed feature distributions, overlapping classes, and redundant information in the feature space.

The observations obtained at this stage influence subsequent decisions regarding feature selection, model evaluation, and metric selection.

2.5 Feature Selection

Feature selection was performed to identify the most informative variables for protein localization prediction. Different feature selection approaches were evaluated.

Since the dataset already contained normalized features and highly unstable multiclass distributions, univariate statistical methods such as chi-square tests were not prioritized, as they could reduce the stability of independent feature ranking methods. Recursive Feature Elimination (RFE) was also evaluated, but it was not chosen as the primary approach.

Instead, feature importance scores generated with Random Forest were used as the primary feature evaluation strategy. Tree-based feature importance methods are more robust to nonlinear relationships, multiclass problems, and feature scaling differences, making them more suitable for multiclass datasets with different feature behaviors (More et al., 2017).

To prevent data leakage, these tests were only performed on the training folds (0, 1, and 2). The independent benchmark fold (-1) was completely removed from the feature selection procedures and reserved only for final evaluation.

The resulting feature importances were used to analyze the relative contribution of different biological features to localization prediction and to support subsequent model development and interpretation.

2.6 Model Training and Evaluation

Various supervised machine learning algorithms were evaluated for the protein localization classification task. Models were selected to represent different learning strategies, including linear, distance-based, probabilistic, and tree-based approaches.

Model development and evaluation were performed using training folds (0, 1, and 2) within a cross-validation framework, while the independent benchmark fold (-1) was reserved solely for final performance evaluation. This separation ensured that model optimization and evaluation remained independent throughout the workflow.

Model performance was assessed using multiple complementary metrics. Accuracy was used as a general measure of overall prediction performance. However, due to the multi-class and unbalanced nature of the dataset, additional metrics including precision, recall, F1 score, and Matthews Correlation Coefficient (MCC) were also considered. Additionally, MCC was important as it provided a more balanced assessment in classification problems involving imbalanced class distributions (Boughorbel et al., 2017).

Classification reports and confusion matrices were also used to examine class-specific performance and misclassification patterns among localization categories. These analyses allowed for a more detailed interpretation of model behavior, beyond general accuracy values.

Finally, with the random forest, which was chosen as the most powerful, the final model was saved for future use.

3. Results

3.1 Exploratory Data Analysis Results

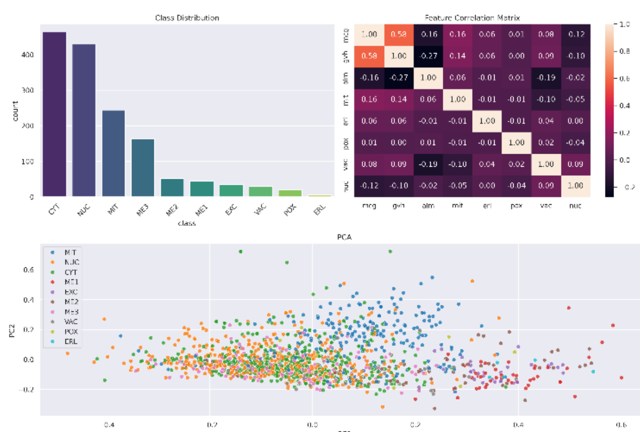

Figure 1. Summary of EDA including class distribution, feature correlation heatmap, and PCA of the Yeast dataset.

As shown in Figure 1, the dataset exhibits significant class imbalance among ten different localization categories. Cytoplasmic (CYT) and nuclear (NUC) proteins are represented by a high number of samples, while endoplasmic reticulum (ERL) and peroxisomal (POX) proteins contain a relatively small number of observations. This indicates that evaluating model performance solely with the accuracy metric can be misleading and necessitates the use of additional performance metrics such as F1-score and MCC.

The PCA projection revealed significant overlap between classes overall, although partial separation was observed between some localization classes. This observation suggests that the dataset is not linearly separable and that the classification problem involves a biologically complex structure.

Correlation analysis showed a generally low level of association between features. However, a moderately positive correlation was observed between mcg and gvh features, both based on signal sequence recognition methods. Overall, exploratory analysis revealed three key challenges of the dataset: multiple class imbalance, overlapping class boundaries, and heterogeneous feature behavior.

Detailed exploratory visualizations, including histograms, density plots, box plots, and skewness analyses, are presented in the supplementary materials (Supplementary Figures S1–S4). These analyses specifically showed that 'ERL' and 'POX' features exhibited strong positive skewness. This points to a biologically significant sparsity resulting from the presence of specific cellular localization signals only in a limited number of protein groups.

3.2 Feature Selection Results

Different feature selection methods were evaluated to identify the most informative features. Due to the small size, multi-class, and imbalanced nature of the dataset, linear and univariate methods would not be sufficient. Therefore, both classical statistical methods and tree-based methods were compared.

First, a Chi-square univariate feature selection method was applied. This method identified mcg, mit, and pox with the highest scores. However, since the method evaluated each feature independently and could be affected by class imbalance in the dataset, the results would not be sufficiently reliable.

Secondly, a Logistic Regression-based Recursive Feature Elimination (RFE) method was applied. RFE results selected mcg, gvh, alm, and mit as the most important features. However, since Logistic Regression is a linear model, it would be limited for a biologically complex problem involving nonlinear relationships, such as protein localization.

More consistent results were obtained when using tree-based methods. Both ExtraTreesClassifier and Random Forest analyses identified alm, mcg, gvh, and mit as the features with the highest significance. It is noteworthy that these features are biologically significant because they represent mechanisms directly related to protein localization, such as membrane region prediction, signal sequence recognition, and mitochondrial targeting, respectively.

In addition, erl and pox features generated lower overall significance scores. However, this does not indicate that these features are biologically insignificant. These features represent rare localization signals active only in specific minority classes, and therefore their global significance scores remained low.

The largely preserved Random Forest significance scores across different folds demonstrated that feature ranking was not dependent on a specific data subset and that the results were stable. Therefore, Random Forest-based feature significance analysis was considered the most reliable feature selection approach in this study.

Although some features had lower significance scores, all original features were preserved during the modeling stage to prevent loss of biologically significant information due to the small sample size.

3.3 Model Evaluation and Comparison

3.3.1 Baseline Logistic Regression Performance

A basic performance evaluation was conducted using Logistic Regression. The model was evaluated with stratified train/test split and stratified 3-fold cross-validation. Stratification was used due to class imbalance.

The train/test split had approximately 55.6% accuracy, also the stratified cross-validation average accuracy was calculated as approximately 54.1% ± 1.4. The similarity of the results indicates that the model performance is generally stable. However, the accuracy values did not adequately represent the complex structure of the dataset of the Logistic Regression model.

Examination of the confusion matrix and classification report revealed that the model performed poorly, particularly in minority classes. The fact that the F1 score was zero in the ERL, EXC, ME2, POX, and VAC classes indicates that the model could not effectively predict these classes. In contrast, higher performance was achieved on the dominant classes, CYT and NUC.

The Matthews Correlation Coefficient (MCC ≈ 0.41) indicates that the model has moderate predictive power. Overall, the results show that the dataset is not linearly separable and that more flexible models are needed.

3.3.2 Comparison of Machine Learning Algorithms

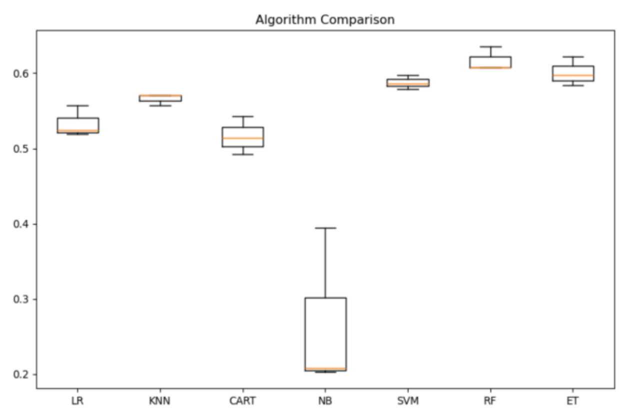

Figure 2. Comparison of cross-validation accuracy scores of different machine learning algorithms.

To compare the performance of different machine learning algorithms, Logistic Regression (LR), k-Nearest Neighbors (KNN), Decision Tree (CART), Gaussian Naive Bayes (NB), Support Vector Machine (SVM), Random Forest (RF), and ExtraTrees (ET) models were evaluated.

The results showed that the highest performance was achieved by the Random Forest model (approximately 61.7% accuracy). ExtraTrees (60.1%) and Support Vector Machine (58.7%) models also showed strong performance. Gaussian Naive Bayes produced the lowest performance.

The fact that tree-based ensemble methods were particularly successful supports the idea that the dataset contains nonlinear relationships. In addition, the Random Forest model produced more stable results across folds. Therefore, the Random Forest approach was preferred as the final model in the subsequent stages.

3.4 Final Model Optimization and Benchmark Evaluation

Following the model comparison results, Random Forest was selected as the most successful approach. After that the hyperparameter optimization was applied using GridSearchCV. Only the training folds (0, 1, 2) were used in the optimization process, while the benchmark fold (-1) was completely excluded.

The best performance was obtained with the following parameters in the grid search:

n_estimators = 300

max_depth = 15

max_features = "log2"

As a result of this optimization, the cross-validation accuracy increased from 61.7% to 62.1%. Although the performance increase was limited, a more stable and optimized model configuration was obtained.

After optimization, the final Random Forest model was retrained on all training folds and evaluated only on the benchmark fold. The final benchmark results were obtained with 62.3% accuracy and MCC ≈ 0.51. Also, the weighted precision ≈ 0.62 and recall ≈ 0.623.

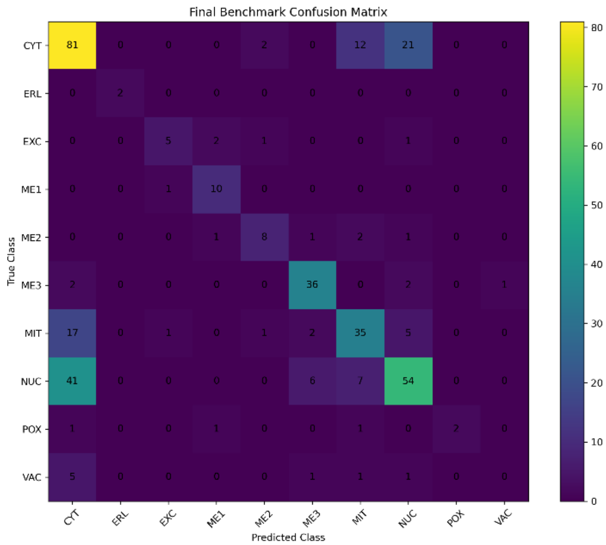

Figure 3. Confusion matrix of the final model on the benchmark dataset.

The confusion matrix and classification report show that the Random Forest model provides significant improvement, especially in minority classes, compared to the basic Logistic Regression model. Successful predictions were obtained in the ERL, EXC, ME1, and POX classes. Also, a significant increase in the macro avg F1-score was observed. This indicates that the model does not focus solely on dominant classes.

However, the VAC class remained one of the most difficult categories to classify. Interestingly, POX and ERL classes were predicted more successfully despite containing very few samples. This suggests that class separability depends not only on the number of samples but also on the discriminative power of biological signals.

One possible biological explanation is that POX and ERL contain more distinct targeting signals, whereas vacuolar proteins (VAC) show more feature overlap with cytoplasmic, membrane-associated, or nuclear proteins. Therefore, the VAC class confounds more strongly with other classes.

Overall, the results show that protein localization is determined by nonlinear and partially overlapping biological mechanisms. Therefore, ensemble-based methods such as Random Forest were able to represent the complex structure of the dataset more successfully compared to linear models.

4. Conclusion

This study aimed to predict the cellular localization regions of yeast proteins using machine learning methods with the Yeast Protein Localization dataset.

Exploratory data analysis show that the dataset contained significant class imbalance, some features exhibited strong skewness, and classes were not linearly separable. Features such as ERL and POX were found to contain biologically significant signals that are active only in specific protein groups.

Different feature selection methods were examined, and tree-based methods were found to be more suitable for the nonlinear structure of the dataset. Model comparisons showed that ensemble-based methods produced the most successful results, and the optimized Random Forest model was selected as the final approach.

The final benchmark evaluation performances are 62% accuracy and MCC ≈ 0.51. The results showed significant improvement, especially in minority classes, according to the basic Logistic Regression model. However, it was observed that some classes, especially VAC, remained challenging due to biological overlaps and limited sample size.

Overall, the study demonstrates that the protein localization problem involves complex and nonlinear biological relationships, and that ensemble-based methods can model this structure more effectively.

Supplementary Materials

Additional figures, exploratory analyses, trained models, and the complete Jupyter notebook used in this project are available in the GitHub repository below:

https://github.com/mbkemec/yeast_protein_localization

Supplementary figures referenced in the report are provided in the `supplementary_materials.pdf` file.

References

Boughorbel, S., Jarray, F., & El-Anbari, M. (2017). Optimal classifier for imbalanced data using Matthews

Correlation Coefficient metric. *PloS one, 12(6), e0177678*.

Botstein, D., & Fink, G. R. (2011). Yeast: an experimental organism for 21st century biology. *Genetics, 189(3), 695-704*.

Hung, M. C., & Link, W. (2011). Protein localization in disease and therapy. *Journal of cell science, 124(20), 3381-3392*.

More, A. S., & Rana, D. P. (2017, October). Review of random forest classification techniques to resolve data imbalance. *In 2017 1st International conference on intelligent systems and information management (ICISIM) (pp. 72-78). IEEE*.

Nakai, K., & Kanehisa, M. (1991). Expert system for predicting protein localization sites in gram-negative bacteria. *PROTEINS: Structure, Function, and Genetics, 11(2), 95–110*.